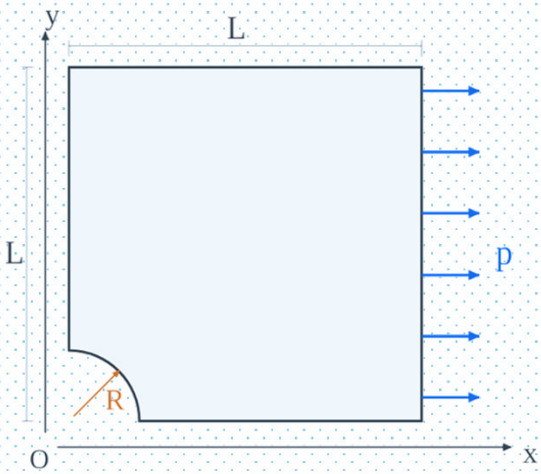
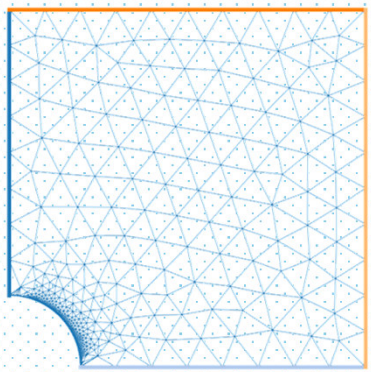
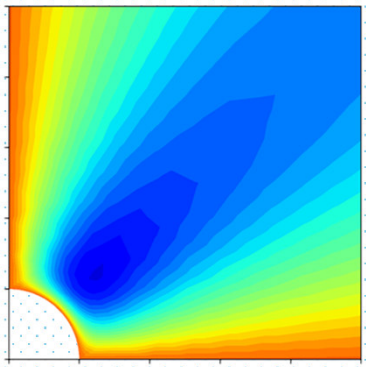
Bauman Moscow State Technical University

Special Topics in Finite Element Modeling for IC engines

Finite Element Analysis of Stress and Strain in a Plate

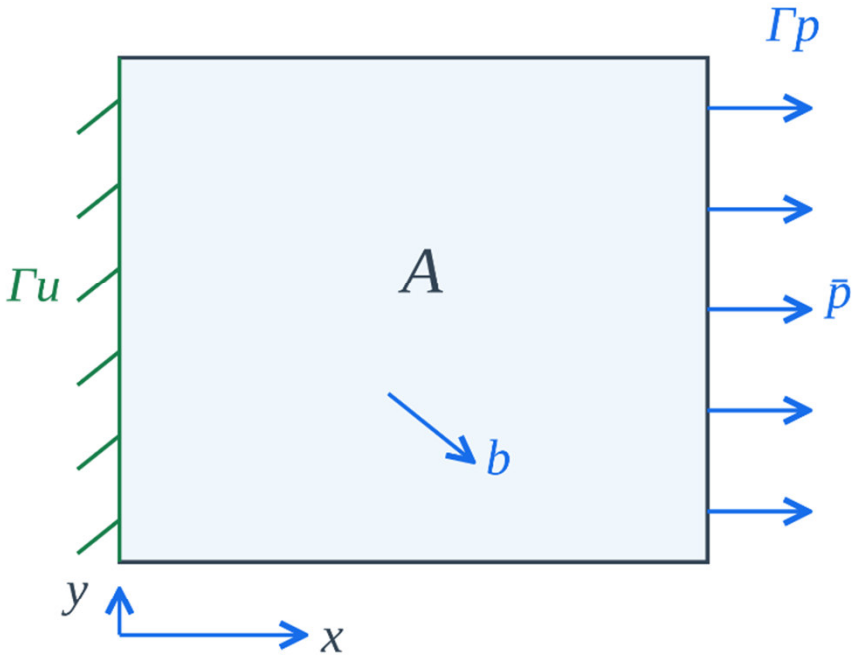






Instructors: Malastowski Nikolay, Blinov Artem

A thin plate subjected to in-plane loads



Schematic geometry and loading.

1. Define the model

Domain A, thickness h, material E and ν.
Prescribe displacements on part of the boundary.
Prescribe tractions on the remaining boundary.
Body forces b may act throughout the domain.

2. Find the fields

$$u(x,y), \quad v(x,y)$$

$$\boldsymbol{\varepsilon}(x,y), \quad \boldsymbol{\sigma}(x,y)$$

We assume small strains, a linear elastic material and plane stress.

Three-dimensional force balance and static equilibrium

1. Force balance in an arbitrary volume

$$\int_V \rho \ddot{\mathbf{u}}\, dV = \int_{\partial V} \boldsymbol{\sigma}\mathbf{n}\, dS + \int_V \mathbf{b}\, dV$$

u = (u, v, w)^T is the displacement vector.
n is the outward unit normal.
b is force per unit volume, N/m³.
ρ is density; ü is acceleration.

2. Local form

$$\rho\ddot{\mathbf{u}} = \nabla\cdot\boldsymbol{\sigma} + \mathbf{b}$$

3. Surface traction

$$\mathbf{p} = \boldsymbol{\sigma}\mathbf{n}$$

$$\boldsymbol{\sigma} = \begin{pmatrix} \sigma_x & \tau_{xy} & \tau_{xz} \\ \tau_{xy} & \sigma_y & \tau_{yz} \\ \tau_{xz} & \tau_{yz} & \sigma_z \end{pmatrix}$$

4. Statics: zero acceleration

$$\boxed{\nabla\cdot\boldsymbol{\sigma} + \mathbf{b} = \mathbf{0}}$$

Equilibrium relates forces. Strain and the constitutive law relate stress to displacement.

Strain–displacement relations in linear elasticity

1. Small displacement gradients

Changes in geometry are neglected in the equilibrium equations.

$$\boldsymbol{\varepsilon} = \frac{1}{2}\left[\nabla \mathbf{u} + (\nabla \mathbf{u})^{\mathsf{T}}\right], \qquad \varepsilon_{ij} = \frac{1}{2}\left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i}\right)$$

2. Components in the plane of the plate

$$\varepsilon_x = \frac{\partial u}{\partial x}, \qquad \varepsilon_y = \frac{\partial v}{\partial y}$$

$$\gamma_{xy} = 2\varepsilon_{xy} = \frac{\partial u}{\partial y} + \frac{\partial v}{\partial x}$$

γxy is the engineering shear strain, twice the tensor component εxy.

Hooke’s law for an isotropic material

1. General linear elastic law

$$\sigma_{ij} = \sum_{k,l \in \{x,y,z\}} C_{ijkl}\varepsilon_{kl}$$

$$i, j, k, l \in \{x, y, z\}$$

C_{ijkl} : stiffness-tensor components, Pa.

Material assumptions

Homogeneous and isotropic.
No initial, plastic
or thermal strains.

2. Elasticity tensor of an isotropic material

$$C_{ijkl} = \lambda_L \, \delta_{ij}\delta_{kl} + \mu \left(\delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk}\right)$$

$$\delta_{ij} = 1 \text{ if } i = j, \qquad \delta_{ij} = 0 \text{ if } i \neq j$$

δ is the Kronecker delta

μ = G is the shear modulus

3. Hooke’s law and two independent elastic constants

$$\boldsymbol{\sigma} = 2G\boldsymbol{\varepsilon} + \lambda_L \, \mathrm{tr}(\boldsymbol{\varepsilon})\mathbf{I}$$

$$\mathrm{tr}(\boldsymbol{\varepsilon}) = \varepsilon_x + \varepsilon_y + \varepsilon_z$$

$$G = \frac{E}{2(1+\nu)}, \qquad \lambda_L = \frac{E\nu}{(1+\nu)(1-2\nu)}$$

tr(ε) : trace of the strain tensor

I : identity tensor

E : Young’s modulus

$\lambda_L = \lambda$: first Lamé parameter

ν : Poisson’s ratio

For E > 0 and −1 < ν < 1/2, strain energy is positive for every nonzero strain.

A thin plate under in-plane loading

1. Mechanical assumptions

The broad faces are traction-free.
Bending is excluded.
Displacements u and v depend on x, y.

$$\sigma_z = \tau_{xz} = \tau_{yz} = 0$$

2. Through-thickness strain remains

$$\varepsilon_z = -\frac{\nu}{E}(\sigma_x + \sigma_y)$$

3. The 3D Hooke law gives

$$\varepsilon_x = \frac{\sigma_x - \nu\sigma_y}{E}$$

$$\varepsilon_y = \frac{\sigma_y - \nu\sigma_x}{E}$$

$$\tau_{xy} = G\gamma_{xy}$$

Solve the first two equations
for σx and σy.

The condition σz = 0 does not imply εz = 0. Plane strain requires a different model.

Constitutive matrix of the plane-stress model

Relation between stress and strain vectors

$$\begin{pmatrix} \sigma_x \\ \sigma_y \\ \tau_{xy} \end{pmatrix} = \underbrace{\frac{E}{1-\nu^2}\begin{pmatrix} 1 & \nu & 0 \\ \nu & 1 & 0 \\ 0 & 0 & \frac{1-\nu}{2} \end{pmatrix}}_{\mathbf{D}} \begin{pmatrix} \varepsilon_x \\ \varepsilon_y \\ \gamma_{xy} \end{pmatrix}$$

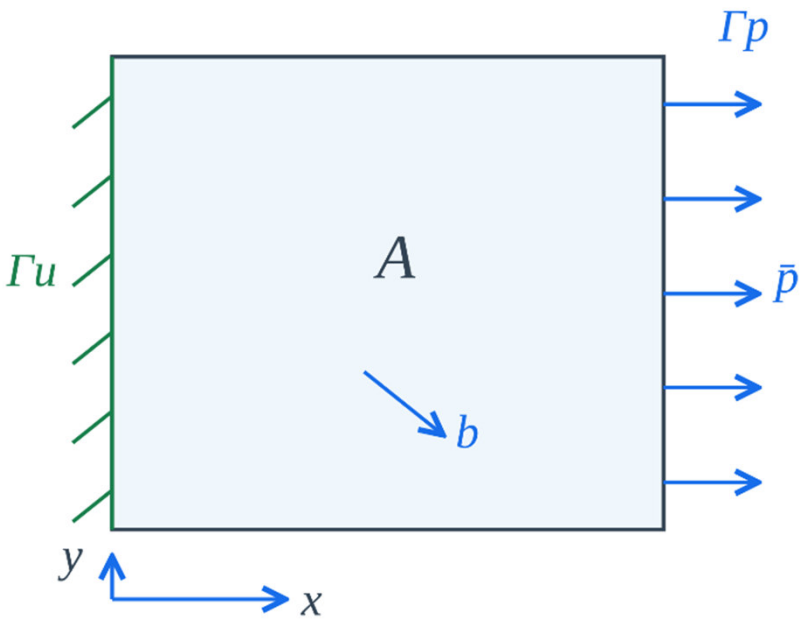
Equilibrium equations in x, y coordinates

$$\frac{\partial \sigma_x}{\partial x} + \frac{\partial \tau_{xy}}{\partial y} + b_x = 0, \qquad \frac{\partial \tau_{xy}}{\partial x} + \frac{\partial \sigma_y}{\partial y} + b_y = 0$$

D is a 3 × 3 matrix. Thickness h does not enter the local constitutive law.

For integration over the planar domain: dV = h dA and dSside = h dℓ.

Boundary conditions for the plane problem



Γu: boundary with prescribed displacements
Γp: boundary with prescribed surface tractions

1. Prescribed displacements

$$\mathbf{u} = \overline{\mathbf{u}} \quad \text{on } \Gamma_u, \qquad \mathbf{u} = \begin{pmatrix} u \\ v \end{pmatrix}$$

2. Prescribed surface tractions

$$\boldsymbol{\sigma}\mathbf{n} = \overline{\mathbf{p}} \quad \text{on } \Gamma_p$$

3. Traction-free boundary

$$\overline{\mathbf{p}} = \mathbf{0} \quad \Longrightarrow \quad \boldsymbol{\sigma}\mathbf{n} = \mathbf{0}$$

n is the outward unit normal to the boundary.

Supports must eliminate rigid-body motion. The prescribed traction p̄ is measured in Pa.

From equilibrium equations to a functional

Total potential energy of the system

1. Associate a number with each field

A functional assigns an energy value to the entire displacement field.

$$\{u(x,y), v(x,y)\} \longmapsto \Pi[u,v]$$

2. Admissible fields

Continuous, with finite energy.
Satisfy prescribed displacements.
Need not yet satisfy equilibrium.

3. Equilibrium minimizes the energy

$$\Pi[u^*, v^*] = \min_{\substack{u,v \\ \text{admissible}}} \Pi[u,v]$$

This holds for a linear elastic body with prescribed forces independent of displacement.

Supports eliminate rigid-body translations and rotation.

FEM seeks the minimum over a finite set of unknown nodal displacements.

Three terms in the total potential energy

1. Strain energy per unit volume

$$W = \frac{1}{2}\left(\sigma_x \varepsilon_x + \sigma_y \varepsilon_y + \tau_{xy}\gamma_{xy}\right)$$

$$W = \frac{E}{2(1-\nu^2)}\left[\varepsilon_x^2 + 2\nu\varepsilon_x\varepsilon_y + \varepsilon_y^2 + \frac{1-\nu}{2}\gamma_{xy}^2\right]$$

2. Energy of a system with constant thickness h

$$\Pi = h\int_A W\,dA - h\int_A (b_x u + b_y v)\,dA - h\int_{\Gamma_p} (\bar{p}_x u + \bar{p}_y v)\,d\ell$$

Strain energy | Work of body forces | Work of surface tractions

A: planar plate domain; Γ_p: loaded boundary; h: thickness; ℓ: boundary coordinate; $d\ell$: length element.

All terms are measured in J. The factor 1/2 applies to strain energy, not to prescribed-force work.

Total potential energy of the plane model

Functional in terms of displacement derivatives

$$\Pi[u,v] = \frac{hE}{2(1-\nu^2)} \int_A \left[\left(\frac{\partial u}{\partial x}\right)^2 + 2\nu \frac{\partial u}{\partial x}\frac{\partial v}{\partial y} + \left(\frac{\partial v}{\partial y}\right)^2 + \frac{1-\nu}{2}\left(\frac{\partial u}{\partial y} + \frac{\partial v}{\partial x}\right)^2 \right] dA$$

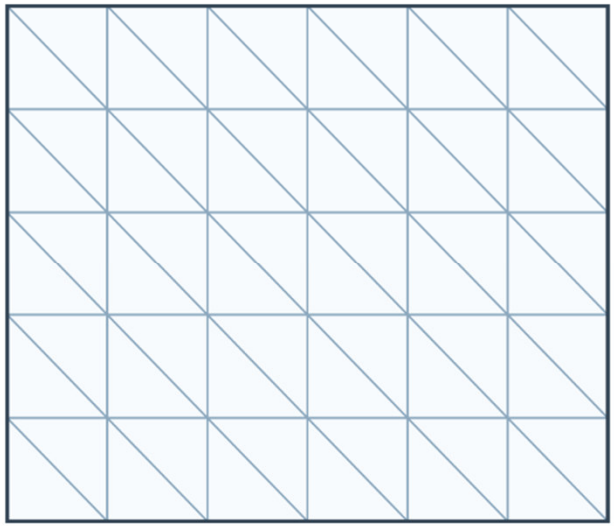
$$- h \int_A (b_x u + b_y v)\, dA - h \int_{\Gamma_p} (\bar{p}_x u + \bar{p}_y v)\, d\ell.$$

Γp denotes boundary segments with prescribed tractions.
Prescribed displacements restrict the admissible fields u, v.

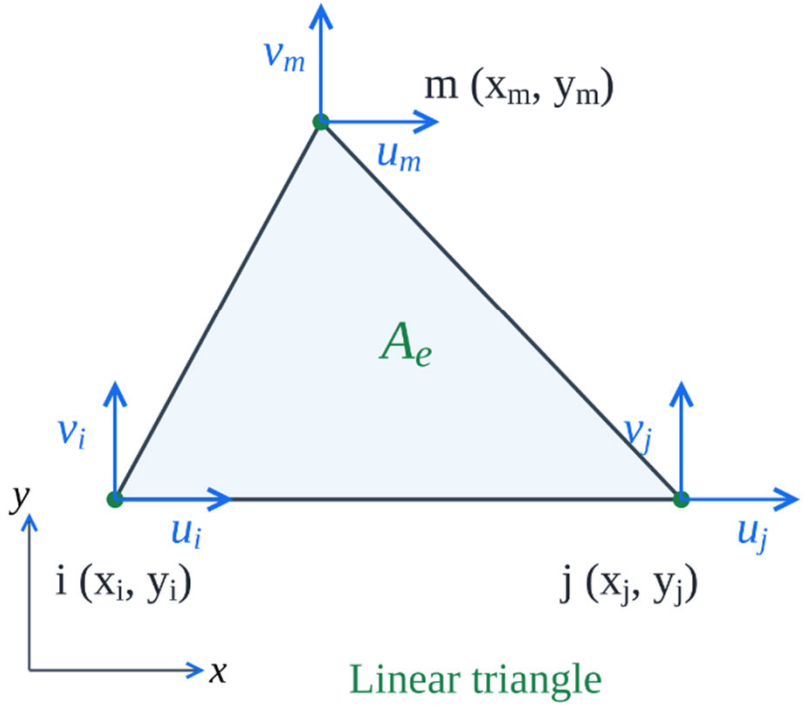
Next, define linear fields u and v within a triangle and substitute them into this functional.

Finite element model of the plate

Divide the domain into triangles and introduce nodal unknowns



Example domain mesh



Linear triangle

Each node has two displacement components: u along x and v along y.

Displacement approximation and element geometry

1. Approximate displacements by linear polynomials

$$u = \beta_0 + \beta_x x + \beta_y y, \qquad v = \eta_0 + \eta_x x + \eta_y y$$

2. The polynomials interpolate the nodal values

$$\begin{pmatrix} u_i \\ u_j \\ u_m \end{pmatrix} = \begin{pmatrix} 1 & x_i & y_i \\ 1 & x_j & y_j \\ 1 & x_m & y_m \end{pmatrix} \begin{pmatrix} \beta_0 \\ \beta_x \\ \beta_y \end{pmatrix} \qquad \begin{pmatrix} v_i \\ v_j \\ v_m \end{pmatrix} = \begin{pmatrix} 1 & x_i & y_i \\ 1 & x_j & y_j \\ 1 & x_m & y_m \end{pmatrix} \begin{pmatrix} \eta_0 \\ \eta_x \\ \eta_y \end{pmatrix}$$

Two displacements at each vertex give six element degrees of freedom.

3. Triangle area and nondegeneracy

$$\Delta_e = \det \begin{pmatrix} 1 & x_i & y_i \\ 1 & x_j & y_j \\ 1 & x_m & y_m \end{pmatrix}$$

$$\Delta_e = (x_j - x_i)(y_m - y_i) - (y_j - y_i)(x_m - x_i)$$

$$A_e = \frac{|\Delta_e|}{2}, \qquad \Delta_e \neq 0$$

The three vertices must not be collinear.

Use the positive area A_e in integration. Retain the sign of Δ_e in the shape functions.

Coefficients of the interpolation polynomials

1. Solve for the polynomial coefficients

$$\begin{pmatrix}\beta_0\\ \beta_x\\ \beta_y\end{pmatrix} = \frac{1}{\Delta_e}\begin{pmatrix}a_i & a_j & a_m\\ b_i & b_j & b_m\\ c_i & c_j & c_m\end{pmatrix}\begin{pmatrix}u_i\\ u_j\\ u_m\end{pmatrix} = \frac{1}{\Delta_e}\begin{pmatrix}a_iu_i + a_ju_j + a_mu_m\\ b_iu_i + b_ju_j + b_mu_m\\ c_iu_i + c_ju_j + c_mu_m\end{pmatrix}$$

$$\begin{pmatrix}\eta_0\\ \eta_x\\ \eta_y\end{pmatrix} = \frac{1}{\Delta_e}\begin{pmatrix}a_i & a_j & a_m\\ b_i & b_j & b_m\\ c_i & c_j & c_m\end{pmatrix}\begin{pmatrix}v_i\\ v_j\\ v_m\end{pmatrix} = \frac{1}{\Delta_e}\begin{pmatrix}a_iv_i + a_jv_j + a_mv_m\\ b_iv_i + b_jv_j + b_mv_m\\ c_iv_i + c_jv_j + c_mv_m\end{pmatrix}$$

2. Write the geometric coefficients of the inverse matrix

$$\begin{aligned} a_i &= x_jy_m - x_my_j, & b_i &= y_j - y_m, & c_i &= x_m - x_j,\\ a_j &= x_my_i - x_iy_m, & b_j &= y_m - y_i, & c_j &= x_i - x_m,\\ a_m &= x_iy_j - x_jy_i, & b_m &= y_i - y_j, & c_m &= x_j - x_i. \end{aligned}$$

The inverse coordinate matrix is the same for u and v: its coefficients depend only on geometry.

Shape functions

1. Substitute the coefficients into the polynomials

$$u = \frac{a_i + b_i x + c_i y}{\Delta_e} u_i + \frac{a_j + b_j x + c_j y}{\Delta_e} u_j + \frac{a_m + b_m x + c_m y}{\Delta_e} u_m$$

$$v = \frac{a_i + b_i x + c_i y}{\Delta_e} v_i + \frac{a_j + b_j x + c_j y}{\Delta_e} v_j + \frac{a_m + b_m x + c_m y}{\Delta_e} v_m$$

Group by nodal displacements: the coefficients of u_a and v_a are identical.

2. Introduce the shape functions

$$N_a(x, y) = \frac{a_a + b_a x + c_a y}{\Delta_e}, \qquad a \in \{i, j, m\}$$

3. Express both fields in the same form

$$u = N_i u_i + N_j u_j + N_m u_m, \qquad v = N_i v_i + N_j v_j + N_m v_m$$

The shape functions N_i, N_j, N_m are the same for u and v and depend only on element geometry.

Shape-function properties and the vector of unknowns

1. Each function equals one at its own node and zero at the others

$$N_i(x_i, y_i) = 1, \qquad N_i(x_j, y_j) = N_i(x_m, y_m) = 0$$

At node i the numerator equals Δ_e; at nodes j and m it equals zero.

2. Collect the six degrees of freedom

$$\mathbf{U}_e = (u_i, v_i, u_j, v_j, u_m, v_m)^{\mathsf{T}}$$

$$\begin{pmatrix} u \\ v \end{pmatrix} = \begin{pmatrix} N_i & 0 & N_j & 0 & N_m & 0 \\ 0 & N_i & 0 & N_j & 0 & N_m \end{pmatrix} \mathbf{U}_e$$

Adjacent elements share nodal values along their common edge, so the field is continuous.

Differentiation of the shape functions

1. Compute the spatial derivatives

$$\frac{\partial N_a}{\partial x} = \frac{b_a}{\Delta_e}, \qquad \frac{\partial N_a}{\partial y} = \frac{c_a}{\Delta_e}, \qquad a \in \{i, j, m\}$$

2. Substitute into the strain–displacement relations

$$\varepsilon_x = \frac{\partial u}{\partial x} = \frac{b_i u_i + b_j u_j + b_m u_m}{\Delta_e}, \qquad \varepsilon_y = \frac{\partial v}{\partial y} = \frac{c_i v_i + c_j v_j + c_m v_m}{\Delta_e}$$

$$\gamma_{xy} = \frac{\partial u}{\partial y} + \frac{\partial v}{\partial x} = \frac{c_i u_i + b_i v_i + c_j u_j + b_j v_j + c_m u_m + b_m v_m}{\Delta_e}$$

Linear shape functions have constant derivatives within the triangle.

Strain–displacement matrix

1. Collect the coefficients of the nodal displacements

$$\boldsymbol{\varepsilon}_e = \begin{pmatrix} \varepsilon_x \\ \varepsilon_y \\ \gamma_{xy} \end{pmatrix} = \mathbf{B}_e \mathbf{U}_e$$

$$\mathbf{B}_e = \frac{1}{\Delta_e} \begin{pmatrix} b_i & 0 & b_j & 0 & b_m & 0 \\ 0 & c_i & 0 & c_j & 0 & c_m \\ c_i & b_i & c_j & b_j & c_m & b_m \end{pmatrix}_{3\times 6}$$

B_e has units of 1/m.
Strains are constant within a linear triangular element.

B_e collects the shape-function derivatives. It is not a separate physical law.

Functional after substituting the fields

1. Strain energy

$$\Pi_e = \frac{hE}{2(1-\nu^2)\Delta_e^2}\int_{A_e}\Big[(b_iu_i+b_ju_j+b_mu_m)^2 + 2\nu(b_iu_i+b_ju_j+b_mu_m)(c_iv_i+c_jv_j+c_mv_m)$$
$$+ (c_iv_i+c_jv_j+c_mv_m)^2 + \frac{1-\nu}{2}(c_iu_i+b_iv_i+c_ju_j+b_jv_j+c_mu_m+b_mv_m)^2\Big]\,dA$$

2. Potential of body forces

$$-h\int_{A_e}\Big[b_x(N_iu_i+N_ju_j+N_mu_m)+b_y(N_iv_i+N_jv_j+N_mv_m)\Big]\,dA$$

3. Potential of boundary forces

$$-h\int_{\Gamma_{p,e}}\Big[\bar{p}_x(N_iu_i+N_ju_j+N_mu_m)+\bar{p}_y(N_iv_i+N_jv_j+N_mv_m)\Big]\,d\ell.$$

First form all energy terms. Differentiate with respect to nodal displacements later.

Integrating the strain energy

1. Constant derivatives make integration straightforward

$$\Pi_{\mathrm{el},e} = \frac{hEA_e}{2(1-\nu^2)\Delta_e^2}\Big[(b_i u_i + b_j u_j + b_m u_m)^2 + (c_i v_i + c_j v_j + c_m v_m)^2$$

$$+ 2\nu(b_i u_i + b_j u_j + b_m u_m)(c_i v_i + c_j v_j + c_m v_m)$$

$$+ \frac{1-\nu}{2}(c_i u_i + b_i v_i + c_j u_j + b_j v_j + c_m u_m + b_m v_m)^2\Big].$$

2. Obtain a quadratic function of nodal displacements

There are no equilibrium equations yet. We have only evaluated the first energy contribution.

For constant E, ν and h, integration over the element gives the area factor A_e.

Expanding the quadratic terms

1. Expand the squares of sums

$$(b_i u_i + b_j u_j + b_m u_m)^2 = b_i^2 u_i^2 + b_j^2 u_j^2 + b_m^2 u_m^2 + 2b_i b_j u_i u_j + 2b_i b_m u_i u_m + 2b_j b_m u_j u_m,$$

$$(c_i v_i + c_j v_j + c_m v_m)^2 = c_i^2 v_i^2 + c_j^2 v_j^2 + c_m^2 v_m^2 + 2c_i c_j v_i v_j + 2c_i c_m v_i v_m + 2c_j c_m v_j v_m.$$

2. Expand the mixed product

$$(b_i u_i + b_j u_j + b_m u_m)(c_i v_i + c_j v_j + c_m v_m) = b_i c_i u_i v_i + b_i c_j u_i v_j + b_i c_m u_i v_m + b_j c_i u_j v_i + b_j c_j u_j v_j + b_j c_m u_j v_m + b_m c_i u_m v_i + b_m c_j u_m v_j + b_m c_m u_m v_m.$$

The coefficients of displacement products will later form the stiffness matrix.

Work of constant body forces

1. Write the functional term

$$\Pi_{b,e} = -h \int_{A_e} \left[b_x (N_i u_i + N_j u_j + N_m u_m) + b_y (N_i v_i + N_j v_j + N_m v_m) \right] dA$$

2. Integrate linear functions in ordinary coordinates

$$\int_{A_e} f(x,y)\, dA = A_e f(x_c, y_c), \quad x_c = \frac{x_i + x_j + x_m}{3}, \quad y_c = \frac{y_i + y_j + y_m}{3}$$

$$N_i(x_c, y_c) = N_j(x_c, y_c) = N_m(x_c, y_c) = \frac{1}{3}, \qquad \int_{A_e} N_i\, dA = \int_{A_e} N_j\, dA = \int_{A_e} N_m\, dA = \frac{A_e}{3}$$

x_c, y_c — centroid coordinates of the triangular finite element

The centroid rule integrates a linear function over a triangle exactly.

Integral of body-force work

1. Take nodal displacements outside the integral

$$\Pi_{b,e} = -h\Big[b_x\left(u_i\int_{A_e} N_i\,dA + u_j\int_{A_e} N_j\,dA + u_m\int_{A_e} N_m\,dA\right) + b_y\left(v_i\int_{A_e} N_i\,dA + v_j\int_{A_e} N_j\,dA + v_m\int_{A_e} N_m\,dA\right)\Big].$$

2. Substitute the integrals $A_e/3$

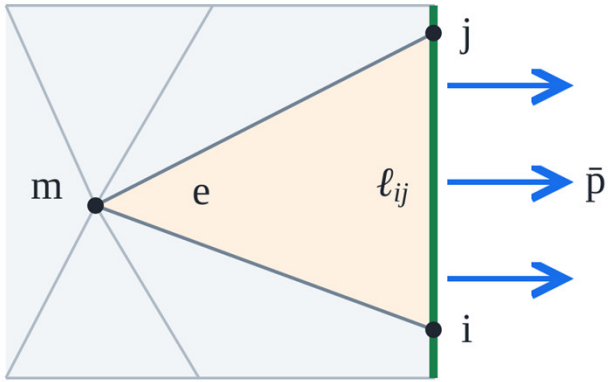
$$\Pi_{b,e} = -\frac{hA_e}{3}\left[b_x(u_i+u_j+u_m) + b_y(v_i+v_j+v_m)\right]$$

If b varies with position, evaluate the integrals of $N_a b$ using its actual spatial variation.

For a constant body force, each node receives one third of the total element force.

Load on a boundary edge

1. Consider a straight edge ij on the outer boundary



$$\ell_{ij} = \sqrt{(x_j - x_i)^2 + (y_j - y_i)^2}$$

$$N_m\big|_{ij} = 0$$

2. Write the work of the load on this edge

$$\Pi^{ij}_{p,e} = -h \int_{ij} [\bar{p}_x (N_i u_i + N_j u_j) + \bar{p}_y (N_i v_i + N_j v_j)]\, d\ell$$

Only an outer boundary edge with prescribed surface traction receives a boundary contribution.

Edge integration in x, y coordinates

1. Use the exact rule for a linear function

$$\int_{ij} f(x,y)\,d\ell = \frac{\ell_{ij}}{2}\left[f(x_i,y_i) + f(x_j,y_j)\right]$$

2. Substitute the endpoint values of the shape functions

$$\int_{ij} N_i\,d\ell = \frac{\ell_{ij}}{2}(1+0) = \frac{\ell_{ij}}{2}, \qquad \int_{ij} N_j\,d\ell = \frac{\ell_{ij}}{2}(0+1) = \frac{\ell_{ij}}{2}$$

3. Obtain the work of a uniform load

$$\Pi_{p,e}^{ij} = -\frac{h\ell_{ij}}{2}\left[\bar{p}_x(u_i+u_j) + \bar{p}_y(v_i+v_j)\right]$$

dℓ denotes physical length. No dimensionless coordinate is needed for this calculation.

Complete scalar element functional

Strain energy

$$\Pi_e = \frac{hEA_e}{2(1-\nu^2)\Delta_e^2}\Big[(b_iu_i+b_ju_j+b_mu_m)^2+(c_iv_i+c_jv_j+c_mv_m)^2$$

$$+2\nu(b_iu_i+b_ju_j+b_mu_m)(c_iv_i+c_jv_j+c_mv_m)+\frac{1-\nu}{2}(c_iu_i+b_iv_i+c_ju_j+b_jv_j+c_mu_m+b_mv_m)^2\Big]$$

Body forces

$$-\frac{hA_e}{3}[b_x(u_i+u_j+u_m)+b_y(v_i+v_j+v_m)]$$

Edge load

$$-\frac{h\ell_{ij}}{2}[\bar{p}_x(u_i+u_j)+\bar{p}_y(v_i+v_j)].$$

Assumptions for this expression

Constant E, ν, h and body forces. Uniform load on one edge ij.
For several loaded edges, add their contributions.

After integration, we obtain an explicit function of six nodal displacements.

Minimization after finite element discretization

1. Sum the energies of all elements

$$\Pi_{\mathrm{FE}}(\mathbf{U}) = \sum_{e=1}^{n_e} \Pi_e(\mathbf{U}_e), \qquad \mathbf{U} = (u_1, v_1, \ldots, u_N, v_N)^{\mathsf{T}}$$

2. Differentiate with respect to an unknown free displacement

$$\frac{\partial \Pi_{\mathrm{FE}}}{\partial U_a} = \sum_{e=1}^{n_e} \frac{\partial \Pi_e}{\partial U_a} = 0$$

a is a global degree-of-freedom index; U_a is the corresponding nodal displacement in x or y.

Elements sharing a node contribute together.
For prescribed displacements, the derivative relates to the support reaction.

The sum of contributions to a free global degree of freedom is zero, not each element contribution.

Strain-energy derivatives with respect to u_a and v_a

1. Restate the strain-energy term

$$\Pi_{\mathrm{el},e} = \frac{hEA_e}{2(1-\nu^2)\Delta_e^2}\Big[(b_iu_i + b_ju_j + b_mu_m)^2 + (c_iv_i + c_jv_j + c_mv_m)^2 + 2\nu(b_iu_i + b_ju_j + b_mu_m)(c_iv_i + c_jv_j + c_mv_m) + \frac{1-\nu}{2}(c_iu_i + b_iv_i + c_ju_j + b_jv_j + c_mu_m + b_mv_m)^2\Big].$$

2. Differentiate with respect to u_a and v_a, where $a \in \{i, j, m\}$

$$\frac{\partial \Pi_{\mathrm{el},e}}{\partial u_a} = \frac{hEA_e}{(1-\nu^2)\Delta_e^2}\Big[b_a(b_iu_i + b_ju_j + b_mu_m) + \nu b_a(c_iv_i + c_jv_j + c_mv_m) + \frac{1-\nu}{2}c_a(c_iu_i + b_iv_i + c_ju_j + b_jv_j + c_mu_m + b_mv_m)\Big].$$

$$\frac{\partial \Pi_{\mathrm{el},e}}{\partial v_a} = \frac{hEA_e}{(1-\nu^2)\Delta_e^2}\Big[c_a(c_iv_i + c_jv_j + c_mv_m) + \nu c_a(b_iu_i + b_ju_j + b_mu_m) + \frac{1-\nu}{2}b_a(c_iu_i + b_iv_i + c_ju_j + b_jv_j + c_mu_m + b_mv_m)\Big].$$

Group by the six nodal displacements: coefficients of u_i, v_i, u_j, v_j, u_m, v_m follow from geometry and material properties. Denote them by K and form the stiffness matrix rows.

Differentiating the square cancels the factor 1/2. The derivatives are linear in nodal displacements.

Stiffness matrix coefficients

For any pair of nodes a, b ∈ {i, j, m}:

$$K^e_{u_a u_b} = \frac{hEA_e}{(1-\nu^2)\Delta_e^2}\left[b_a b_b + \frac{1-\nu}{2} c_a c_b\right],$$

$$K^e_{u_a v_b} = \frac{hEA_e}{(1-\nu^2)\Delta_e^2}\left[\nu b_a c_b + \frac{1-\nu}{2} c_a b_b\right],$$

$$K^e_{v_a u_b} = \frac{hEA_e}{(1-\nu^2)\Delta_e^2}\left[\nu c_a b_b + \frac{1-\nu}{2} b_a c_b\right],$$

$$K^e_{v_a v_b} = \frac{hEA_e}{(1-\nu^2)\Delta_e^2}\left[c_a c_b + \frac{1-\nu}{2} b_a b_b\right].$$

Each K relates a nodal displacement to a nodal force contribution. Units of K: N/m.

From scalar derivatives to matrix form

1. Write derivatives for all six displacements

$$\frac{\partial \Pi_{\mathrm{el},e}}{\partial u_i} = K^e_{u_iu_i}u_i + K^e_{u_iv_i}v_i + K^e_{u_iu_j}u_j + K^e_{u_iv_j}v_j + K^e_{u_iu_m}u_m + K^e_{u_iv_m}v_m$$
$$\frac{\partial \Pi_{\mathrm{el},e}}{\partial v_i} = K^e_{v_iu_i}u_i + K^e_{v_iv_i}v_i + K^e_{v_iu_j}u_j + K^e_{v_iv_j}v_j + K^e_{v_iu_m}u_m + K^e_{v_iv_m}v_m$$
$$\frac{\partial \Pi_{\mathrm{el},e}}{\partial u_j} = K^e_{u_ju_i}u_i + K^e_{u_jv_i}v_i + K^e_{u_ju_j}u_j + K^e_{u_jv_j}v_j + K^e_{u_ju_m}u_m + K^e_{u_jv_m}v_m$$
$$\frac{\partial \Pi_{\mathrm{el},e}}{\partial v_j} = K^e_{v_ju_i}u_i + K^e_{v_jv_i}v_i + K^e_{v_ju_j}u_j + K^e_{v_jv_j}v_j + K^e_{v_ju_m}u_m + K^e_{v_jv_m}v_m$$
$$\frac{\partial \Pi_{\mathrm{el},e}}{\partial u_m} = K^e_{u_mu_i}u_i + K^e_{u_mv_i}v_i + K^e_{u_mu_j}u_j + K^e_{u_mv_j}v_j + K^e_{u_mu_m}u_m + K^e_{u_mv_m}v_m$$
$$\frac{\partial \Pi_{\mathrm{el},e}}{\partial v_m} = K^e_{v_mu_i}u_i + K^e_{v_mv_i}v_i + K^e_{v_mu_j}u_j + K^e_{v_mv_j}v_j + K^e_{v_mu_m}u_m + K^e_{v_mv_m}v_m$$

2. Combine the six equations in matrix form

$$\frac{\partial \Pi_{\mathrm{el},e}}{\partial \mathbf{U}_e} = \mathbf{K}_e\mathbf{U}_e, \qquad \mathbf{K}_e = \mathbf{K}_e^{\mathsf{T}}$$

Three nodes with two components each give six rows and six columns.

The stiffness matrix contains the coefficients of the linear expressions already obtained.

Stiffness matrix of a triangular element

$$\mathbf{K}_e = \begin{pmatrix} K_{u_iu_i} & K_{u_iv_i} & K_{u_iu_j} & K_{u_iv_j} & K_{u_iu_m} & K_{u_iv_m} \\ K_{v_iu_i} & K_{v_iv_i} & K_{v_iu_j} & K_{v_iv_j} & K_{v_iu_m} & K_{v_iv_m} \\ K_{u_ju_i} & K_{u_jv_i} & K_{u_ju_j} & K_{u_jv_j} & K_{u_ju_m} & K_{u_jv_m} \\ K_{v_ju_i} & K_{v_jv_i} & K_{v_ju_j} & K_{v_jv_j} & K_{v_ju_m} & K_{v_jv_m} \\ K_{u_mu_i} & K_{u_mv_i} & K_{u_mu_j} & K_{u_mv_j} & K_{u_mu_m} & K_{u_mv_m} \\ K_{v_mu_i} & K_{v_mv_i} & K_{v_mu_j} & K_{v_mv_j} & K_{v_mu_m} & K_{v_mv_m} \end{pmatrix}_{6\times 6}$$

The index e is omitted from coefficients inside the matrix for clarity.

$$\mathbf{K}_e = hA_e\mathbf{B}_e^{\mathsf{T}}\mathbf{D}\mathbf{B}_e, \qquad \Pi_{\mathrm{el},e} = \frac{1}{2}\mathbf{U}_e^{\mathsf{T}}\mathbf{K}_e\mathbf{U}_e$$

The compact form $B_e^TDB_e$ combines the same coefficients obtained by differentiation.

Derivatives of body-force and boundary-force work

Body forces: contributions from all three nodes

$$\Pi_{b,e} = -\frac{hA_e}{3}\left[b_x(u_i + u_j + u_m) + b_y(v_i + v_j + v_m)\right]$$

$$\frac{\partial \Pi_{b,e}}{\partial u_a} = -\frac{hA_e}{3} b_x, \qquad \frac{\partial \Pi_{b,e}}{\partial v_a} = -\frac{hA_e}{3} b_y, \qquad a \in \{i, j, m\}$$

$$\frac{\partial \Pi_{b,e}}{\partial \mathbf{U}_e} = -\mathbf{F}_{b,e} = -\frac{hA_e}{3}\begin{pmatrix} b_x & b_y & b_x & b_y & b_x & b_y \end{pmatrix}^{\mathsf{T}}$$

Boundary load on edge ij: only nodes i and j contribute

$$\Pi^{ij}_{p,e} = -\frac{h\ell_{ij}}{2}\left[\bar{p}_x(u_i + u_j) + \bar{p}_y(v_i + v_j)\right]$$

$$\frac{\partial \Pi^{ij}_{p,e}}{\partial \mathbf{U}_e} = -\mathbf{F}^{ij}_{p,e} = -\frac{h\ell_{ij}}{2}\begin{pmatrix} \bar{p}_x & \bar{p}_y & \bar{p}_x & \bar{p}_y & 0 & 0 \end{pmatrix}^{\mathsf{T}}$$

The last two zeros correspond to vertex m, which is not on the loaded edge.

Linear terms give constant nodal forces. Units of F components: N.

Full derivative of the element functional

1. Add the derivatives of all terms

$$\frac{\partial \Pi_e}{\partial \mathbf{U}_e} = \mathbf{K}_e \begin{pmatrix} u_i \\ v_i \\ u_j \\ v_j \\ u_m \\ v_m \end{pmatrix} - \frac{hA_e}{3} \begin{pmatrix} b_x \\ b_y \\ b_x \\ b_y \\ b_x \\ b_y \end{pmatrix} - \frac{h\ell_{ij}}{2} \begin{pmatrix} \bar{p}_x \\ \bar{p}_y \\ \bar{p}_x \\ \bar{p}_y \\ 0 \\ 0 \end{pmatrix}$$

2. Write in compact form

$$\frac{\partial \Pi_e}{\partial \mathbf{U}_e} = \mathbf{K}_e \mathbf{U}_e - \mathbf{F}_{b,e} - \mathbf{F}_{p,e} = \mathbf{r}_e$$

Do not set the element residual r_e to zero separately. Next, assemble the entire system.

System assembly: global displacement vector

1. Two degrees of freedom at each node

$$\underbrace{\mathbf{U}}_{2N\times 1} = \begin{pmatrix} u_1 & v_1 & \cdots & u_i & v_i & \cdots & u_N & v_N \end{pmatrix}^{\mathsf{T}}$$

N is the number of mesh nodes. Adjacent elements share the same u and v at a common node.

2. Sum energies and their derivatives

$$\Pi_{\mathrm{FE}}(\mathbf{U}) = \sum_{e=1}^{n_e} \Pi_e(\mathbf{U}_e)$$

$$\frac{\partial \Pi_{\mathrm{FE}}}{\partial \mathbf{U}} = \sum_{e=1}^{n_e} \frac{\partial \Pi_e}{\partial \mathbf{U}} = \mathbf{K}\mathbf{U} - \mathbf{F} = \mathbf{R}$$

First add the element contributions. Then impose equilibrium on the free components.

For free degrees of freedom, Rf = 0. At restrained components, the residual gives reactions.

System assembly: global stiffness matrix

1. Expand the matrix of element (i, j, m) with zero blocks

$$\hat{\mathbf{K}}_e = \begin{pmatrix} \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{K}^e_{ii} & \mathbf{0} & \mathbf{K}^e_{ij} & \mathbf{0} & \mathbf{K}^e_{im} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{K}^e_{ji} & \mathbf{0} & \mathbf{K}^e_{jj} & \mathbf{0} & \mathbf{K}^e_{jm} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{K}^e_{mi} & \mathbf{0} & \mathbf{K}^e_{mj} & \mathbf{0} & \mathbf{K}^e_{mm} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} \end{pmatrix}_{2N \times 2N}$$

2. Sum the contributions

$$\mathbf{K} = \sum_e \hat{\mathbf{K}}_e$$

The nodal block $K_{a\beta}{}^e$ has size 2 × 2.

Zero blocks represent the other groups of nodes.

$$\mathbf{K}_{ii} = \sum_{e \ni i} \mathbf{K}^e_{ii}, \qquad \mathbf{K}_{ij} = \sum_{e \supset \{i,j\}} \mathbf{K}^e_{ij}$$

Only elements containing both nodes i and j contribute to block Kij.

System assembly: global force vector

1. Place nodal forces in their global positions

$$\mathbf{F} = \sum_e \begin{pmatrix} \mathbf{0} \\ \mathbf{F}_i^e \\ \mathbf{0} \\ \mathbf{F}_j^e \\ \mathbf{0} \\ \mathbf{F}_m^e \\ \mathbf{0} \end{pmatrix}_{2N\times 1}$$

$$\mathbf{F}_a^e = \begin{pmatrix} F_{ax}^e \\ F_{ay}^e \end{pmatrix}$$

Each node receives two force components.
Zero blocks correspond to the other nodes.

$$\mathbf{F}_i = \sum_{e \ni i} \mathbf{F}_i^e$$

2. Dimensions of the assembled system

$$\underbrace{\mathbf{K}}_{2N\times 2N}\ \underbrace{\mathbf{U}}_{2N\times 1} - \underbrace{\mathbf{F}}_{2N\times 1} = \underbrace{\mathbf{R}}_{2N\times 1}$$

The global force vector includes all contributions from loaded elements and edges.

Prescribed displacements in the global system

1. Separate free and prescribed components

$$\begin{pmatrix} \mathbf{K}_{ff} & \mathbf{K}_{fc} \\ \mathbf{K}_{cf} & \mathbf{K}_{cc} \end{pmatrix} \begin{pmatrix} \mathbf{U}_f \\ \overline{\mathbf{U}}_c \end{pmatrix} = \begin{pmatrix} \mathbf{F}_f \\ \mathbf{F}_c + \mathbf{R}_c \end{pmatrix}$$

f denotes free degrees of freedom; c denotes components with prescribed displacements.

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2. Solve only the free equations

$$\underbrace{\mathbf{K}_{ff}}_{n_f \times n_f} \underbrace{\mathbf{U}_f}_{n_f \times 1} = \underbrace{\mathbf{F}_f - \mathbf{K}_{fc}\overline{\mathbf{U}}_c}_{n_f \times 1}$$

$$n_f + n_c = 2N$$

Solve the linear system for Uf. Then combine the computed Uf and known Uc into the full vector U.

Recovery of displacements, strains and stresses

1. Extract the element nodal displacements from the global solution

$$\mathbf{U}_e = \begin{pmatrix} u_i & v_i & u_j & v_j & u_m & v_m \end{pmatrix}^{\mathsf{T}}$$

2. Recover the displacement field inside the element

$$u(x,y) = N_i(x,y)u_i + N_j(x,y)u_j + N_m(x,y)u_m$$

$$v(x,y) = N_i(x,y)v_i + N_j(x,y)v_j + N_m(x,y)v_m$$

N_i, N_j, N_m are the shape functions. Shared nodal values ensure displacement continuity.

3. Compute strains and stresses from the displacements

$$\boldsymbol{\varepsilon}_e = \mathbf{B}_e\mathbf{U}_e, \qquad \boldsymbol{\sigma}_e = \mathbf{D}\boldsymbol{\varepsilon}_e = \mathbf{D}\mathbf{B}_e\mathbf{U}_e$$

For the constant-strain triangle (CST), B_e is constant. If D is constant, strains and stresses are constant within the element.

The nodal solution determines displacement, strain and stress fields throughout the domain.

General method: solution, recovery and verification

1. Assemble the system and impose prescribed displacements

$$\Pi_e(\mathbf{U}_e) \longrightarrow \mathbf{K}_e, \mathbf{F}_e \longrightarrow \mathbf{K}, \mathbf{F}$$

$$\mathbf{K}_{ff}\mathbf{U}_f = \mathbf{F}_f - \mathbf{K}_{fc}\overline{\mathbf{U}}_c$$

2. Recover fields and reactions

$$\mathbf{U} \longrightarrow (u, v) \longrightarrow \boldsymbol{\varepsilon}_e = \mathbf{B}_e\mathbf{U}_e \longrightarrow \boldsymbol{\sigma}_e = \mathbf{D}\boldsymbol{\varepsilon}_e$$

$$\mathbf{R}_c = \mathbf{K}_{cf}\mathbf{U}_f + \mathbf{K}_{cc}\overline{\mathbf{U}}_c - \mathbf{F}_c$$

3. Check the solution and discretization accuracy

Free-equation residual, force and reaction balance, units.
Mesh convergence separately checks field accuracy.

The general procedure is complete: solve for displacements, then recover strains and stresses.



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Special Topics in Finite Element Modeling for IC engines

CONCLUSION